higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## COMMUNICATION ELECTRONICS N4

**18 November 2019**

**This marking guideline consists of 9 pages.**

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## QUESTION 1

1.1 False
1.2 True
1.3 False
1.4 False
1.5 True
1.6 True
1.7 False
1.8 True
1.9 False
1.10 False

(10 × 1) **[10]**

## QUESTION 2

2.1 F
2.2 M
2.3 H
2.4 A
2.5 C
2.6 E
2.7 B
2.8 G
2.9 I
2.10 K

(10 × 1) **[10]**

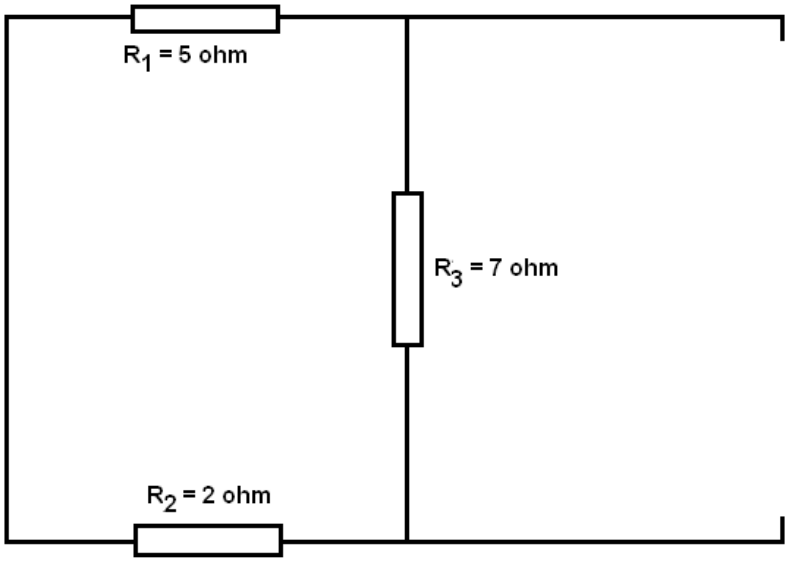
## QUESTION 3: NETWORK THEOREMS

3.1 3.1.1 Kirchhoff's voltage law
In any closed circuit, the algebraic sum of the products of current and resistance of each part of the circuit is equal to the resultant EMF in the circuit.

3.1.2 Norton's theorem
Any two terminal networks of generators and impedances may be replaced by a single constant current generator $I_{N,}$ in parallel, with a single Norton's resistance $R_N$.

(2 × 2) (4)

3.2 Remove the load and the short-circuit battery.

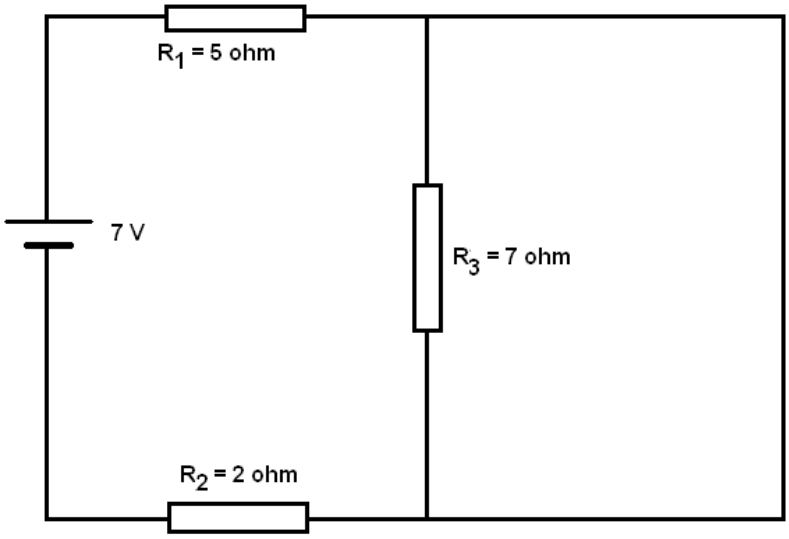



$$R_{TH} = R_{AB} = \frac{R_7 x R_{Ser}}{R_7 + R_{Ser}}$$

$$= \frac{7(5+2)}{7+5+2} \checkmark$$

**= 3,5 Ω✓** (2)

Replace the battery and short-circuit AB.




$R_T = R_1 + R_2$

$= 5 + 2$

**$= 7\ \Omega$✓** (1)

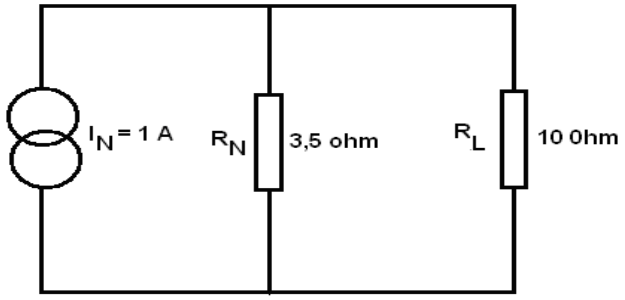
IN $= \frac{V}{R_T}$

$= \frac{7}{7}$

(1)

**= 1 A✓**

Norton's equivalent circuit✓✓




(2)
**[10]**

## QUESTION 4: ALTERNATING CURRENT THEORY

4.1 $X_L = X_C$

$$2\pi fL = \frac{1}{2\pi fc} \checkmark$$

$1 = (2\pi fL) \times (2\pi fc)$

$1 = 4\pi^2 f^2 Lc \checkmark$

$$C = \frac{1}{4\pi^2 f^2 L} \checkmark$$ (3)

4.2 $C = 1/(2\pi f)^2 L$

$= 1/(2\pi \times 85 \times 10^3)^2 \times 10 \times 10^{-3}$

**= 350,6 pF✓** (1)

4.3 4.3.1 B-W = $R/(2\pi L)$

$= 15/(2\pi \times 10 \times 10^{-3})$

**= 238,73 Hz✓**

4.3.2 Q-factor = $X_L/R$

$= 2\pi fL/R$

$= 2\pi \times 85 \times 10^{3\,x}\, 10 \times 10^{-3}/15$

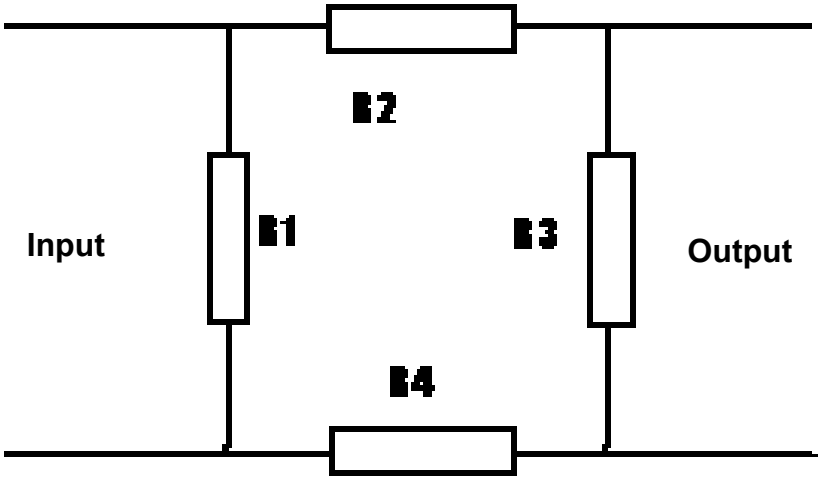
**= 35,6✓**

4.3.3 At resonance Z = R **= 5 Ω**

(3 × 1) (3)
**[7]**

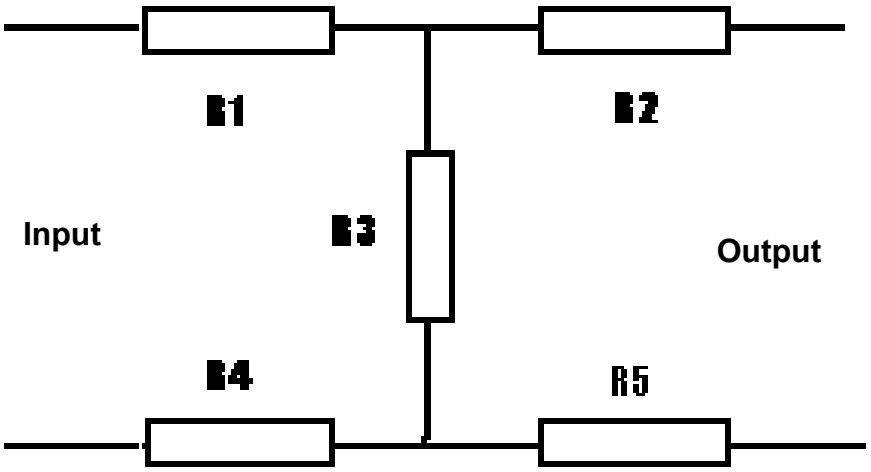
## QUESTION 5: ATTENUATORS AND FILTERS

5.1 5.1.1 **O-TYPE**




✓

5.1.2 **H-TYPE**

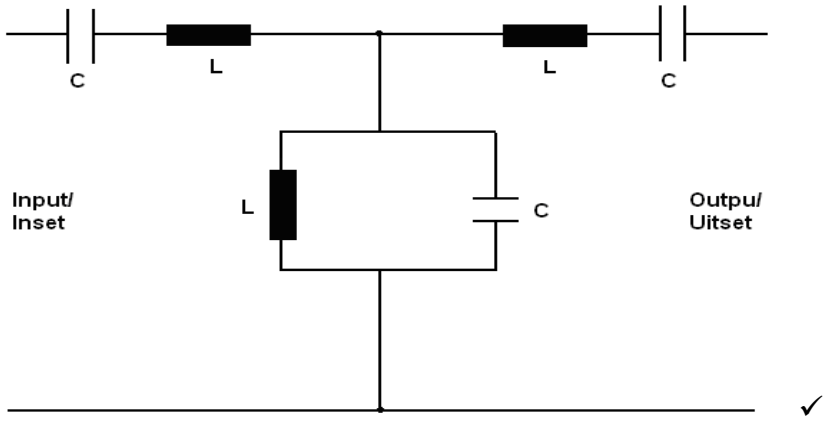



✓

(2 × 1) (2)

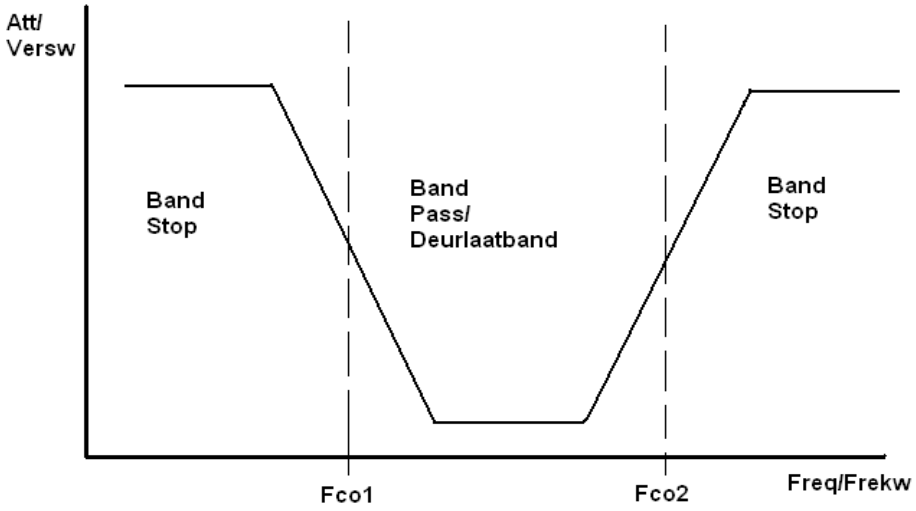
5.2 5.2.1 **BAND-PASS FILTER**

(1)




✓

5.2.2 **ATTENUATION VERSUS FREQUENCY GRAPH FOR A BAND-PASS FILTER**




(2)
**[5]**

## QUESTION 6: MODULATION

6.1 6.1.1 Frequency modulation: It is a process whereby the frequency of the carrier wave is made to vary in accordance with the instantaneous value of the modulating signal.

6.1.2 Modulation index: It is the frequency deviation ratio and the modulating frequency or the degree of modulation in an FM system.

(2 × 2) (4)

6.2 6.2.1 The modulation process is whereby a radio frequency, amplitude or phase is changed in order to transmit the modulating signal or intelligence.

6.2.2 The radio communication process is made possible by the use of electrical energy that travels from the transmitter to the receiver in the form of radio waves.

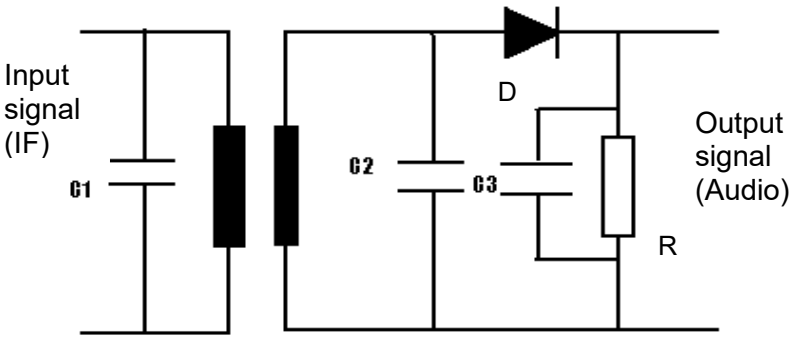
(2 × 2) (4)

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**[8]**

## QUESTION 7: DEMODULATION

7.1 **AM HALF-WAVE DETECTOR**




(3)

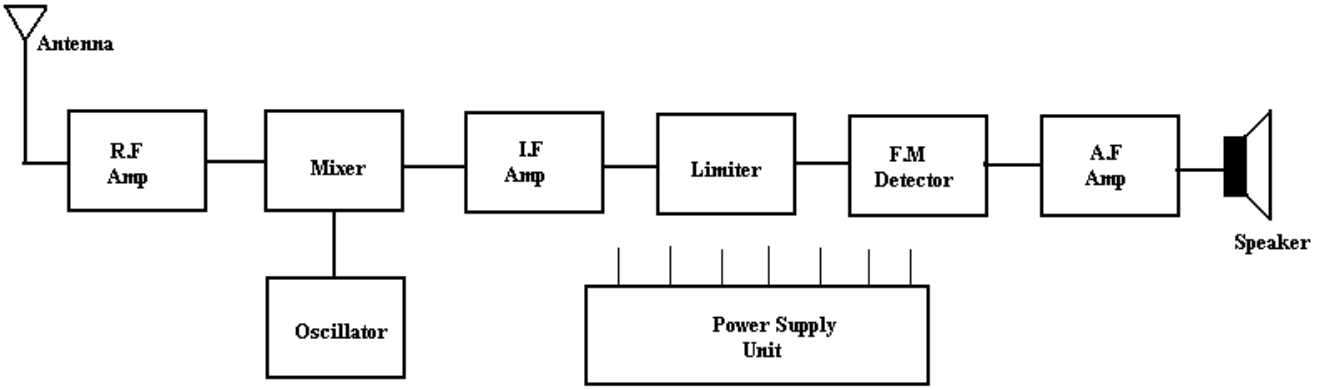
7.2
- During the positive half-cycles of the modulated signal, capacitor C2 will charge to the peak value of the modulating signal through diode D.
- During the negative half-cycles of the modulated signal, capacitor C2 will discharge across resistor R,
- producing a volt drop in accordance with the modulating signal.
- The primary winding and the secondary winding of the transformer and capacitor C2 are tuned to the incoming signal.
- Diode D rectifies the modulating signal and
- capacitor C3 filters off the radio frequency and
- resistor R produces a voltage drop proportional to the audio-frequency modulated wave.

(7)
**[10]**

## QUESTION 8: COMMUNICATION SYSTEMS

8.1

**FM SUPERHETERODYNE RECEIVER**




(5)

8.2
- Antenna: Converts electromagnetic waves into minute electrical currents.
- RF amplifier: Uses tuned circuits to select the desired RF carrier signal.
- Mixer: Combines the incoming RF with the frequency of the local oscillator.
- Oscillator: Generates the signal of the constant amplitude at a fixed difference frequency at above or below the incoming RF.
- IF amplifier: Amplifies the intermediate frequency and gives a high degree of stability to the IF signal.
- Limiter: Prevents amplitude variations by clipping the top and bottom of the RF wave.
- FM detector: Changes frequency variations into audio signals for amplification.
- AF amplifier: Amplifies the audio output from the detector to drive the loudspeaker.
- Loudspeaker: It converts audio frequency signals into sound waves.
- Power supply unit: It supplies regulated DC voltage to all stages. (10 × 1) (10)

**[15]**

## QUESTION 9: RADIO WAVE PROPAGATION

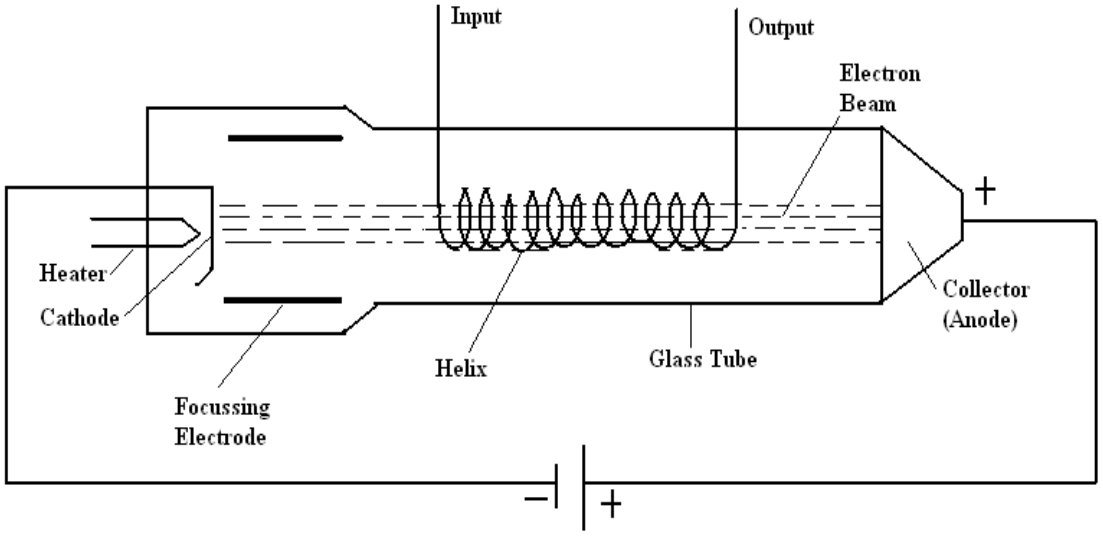
9.1 9.1.1 Silent zone: The silent region extends from the limit of the useful ground wave to the distant point where the signal is first heard.

9.1.2 Critical frequency: It is the maximum frequency at which a radio wave directed vertically upwards is refracted by a layer of the ionosphere and above which it penetrates the layer.

9.1.3 Polarisation: It refers to the direction space of the electric field which is radiated from the antenna and is parallel to the antenna itself.

(3 × 3) (9)

9.2 **TRAVELLING-WAVE TUBE**


Input
Output
Electron
Beam
+
Heater
Cathode
Collector
(Anode)
Helix
Glass Tube
Focussing
Electrode
–
+

(3)

9.3 OPERATION OF THE TRAVELLING-WAVE TUBE

The heater heats the cathode.

- The cathode emits electrons.
- The electron gun emits a narrow beam of electrons with acceleration proportional to anode voltage to be measured in kilovolts.
- Focusing electrodes focus a beam of electrons through a helix at a collector electrode and holds the beam in the centre of the helix.

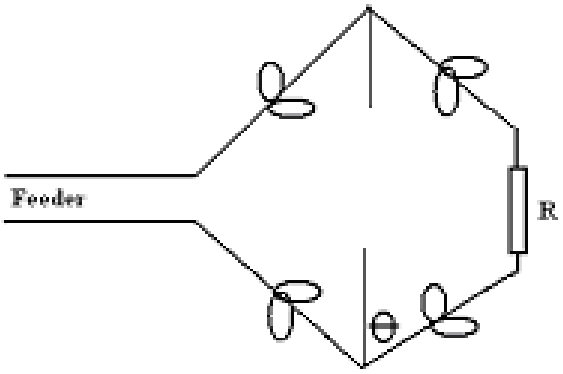
OPERATION OF SPECIFIC ELEMENTS

- The tube makes use of interaction between an electron beam and a travelling-wave signal.
- The RF signal to be amplified is coupled to the input end of the helix to force the electrons in the beam to slow down and to speed up.
- Bunches of electrons are formed and add energy to the RF signal. The signal wave proceeds towards the output end of the helix with its amplitude increasing exponentially. (7)

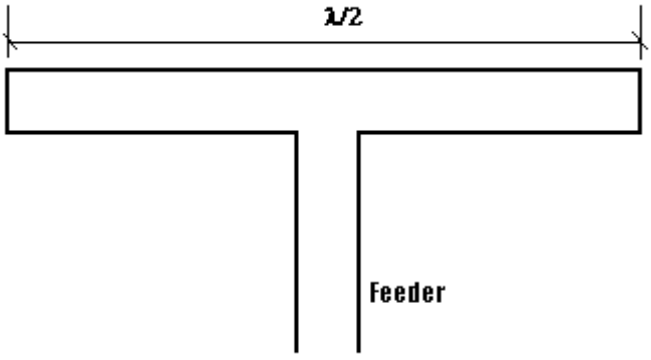
9.4
- A directional antenna concentrates radiation mainly in one direction.
- With an omnidirectional antenna radiation is spread in all directions.

(2 × 1) (2)

9.5 9.5.1 Rhombic antenna




9.5.2 Folded dipole antenna




(2 × 2) (4)

**[25]**

**TOTAL: 100**

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